

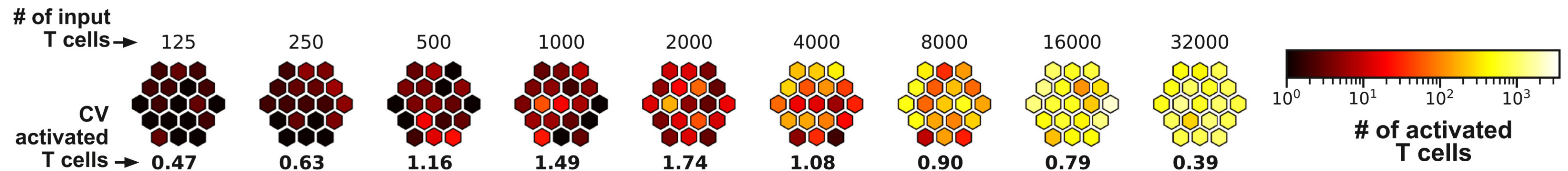
Mapping the cell fate landscape with optimal transport

From the Waddington landscape to seascape

Motivation

Stochasticity in cancer immunotherapies

- Why do cancer immunotherapies have high patient-level variation ?
- Experiment : seed wells with melanoma tumor cells and increasing numbers of cytotoxic T-cells => observe the variations.
- Measure cell protein expression with flow cytometry
- At low numbers, variations correlate with number of input T-cells.



“Spark” cells drive the immune response by producing IFN- γ

Successful therapies should use spark cells

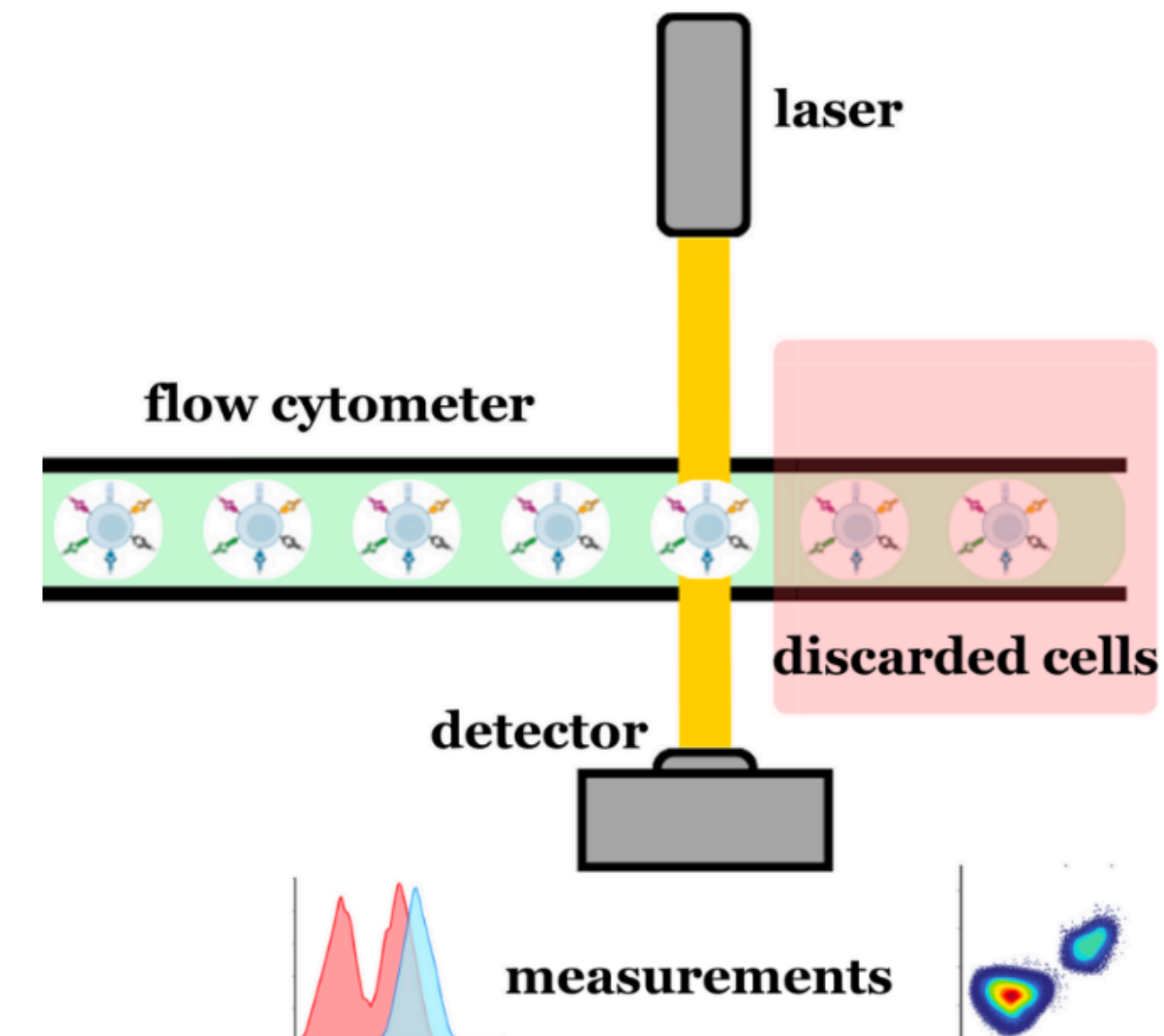
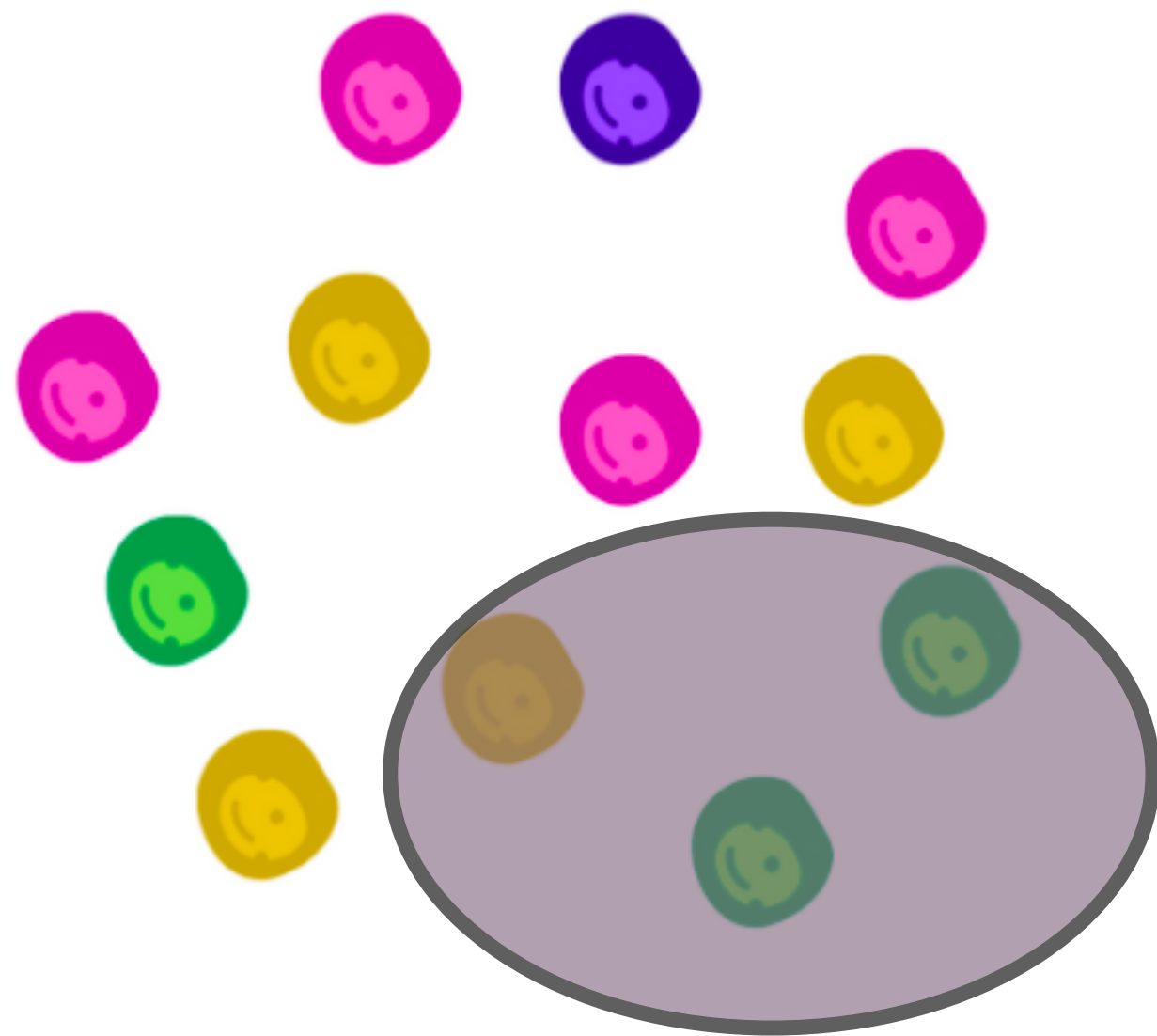
- Stochastic model reveals importance of spark cells to the immune response.
- Knowing which cells are poised to transition to spark increases therapeutic efficacy and efficiency.



Background

Measurement problem - experimental techniques are destructive

- Cells are measured and not replaced
- We want to infer cell trajectories from population data => this is a hard problem



Background

Optimal transport (static)

- Find a joint distribution π such that

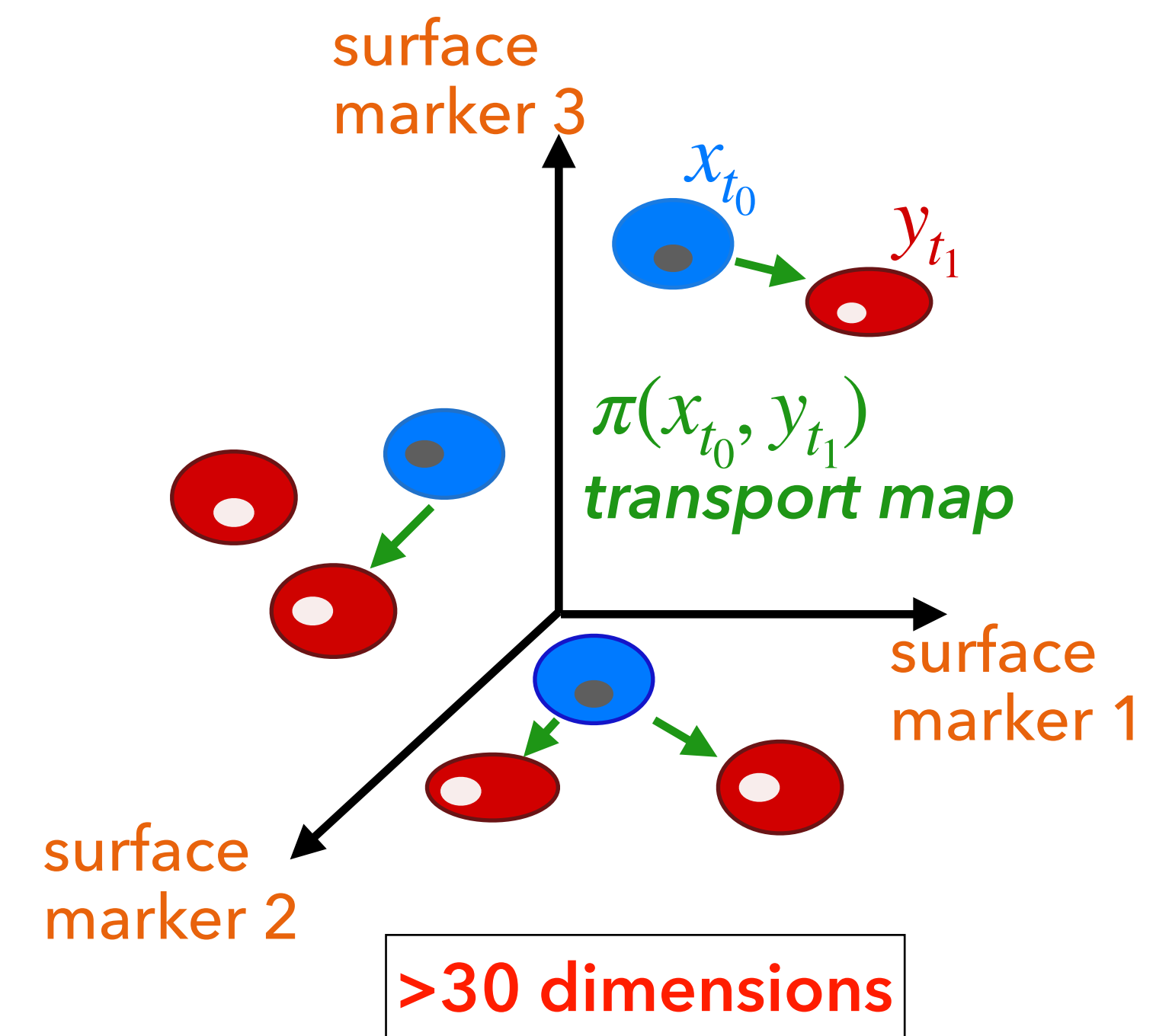
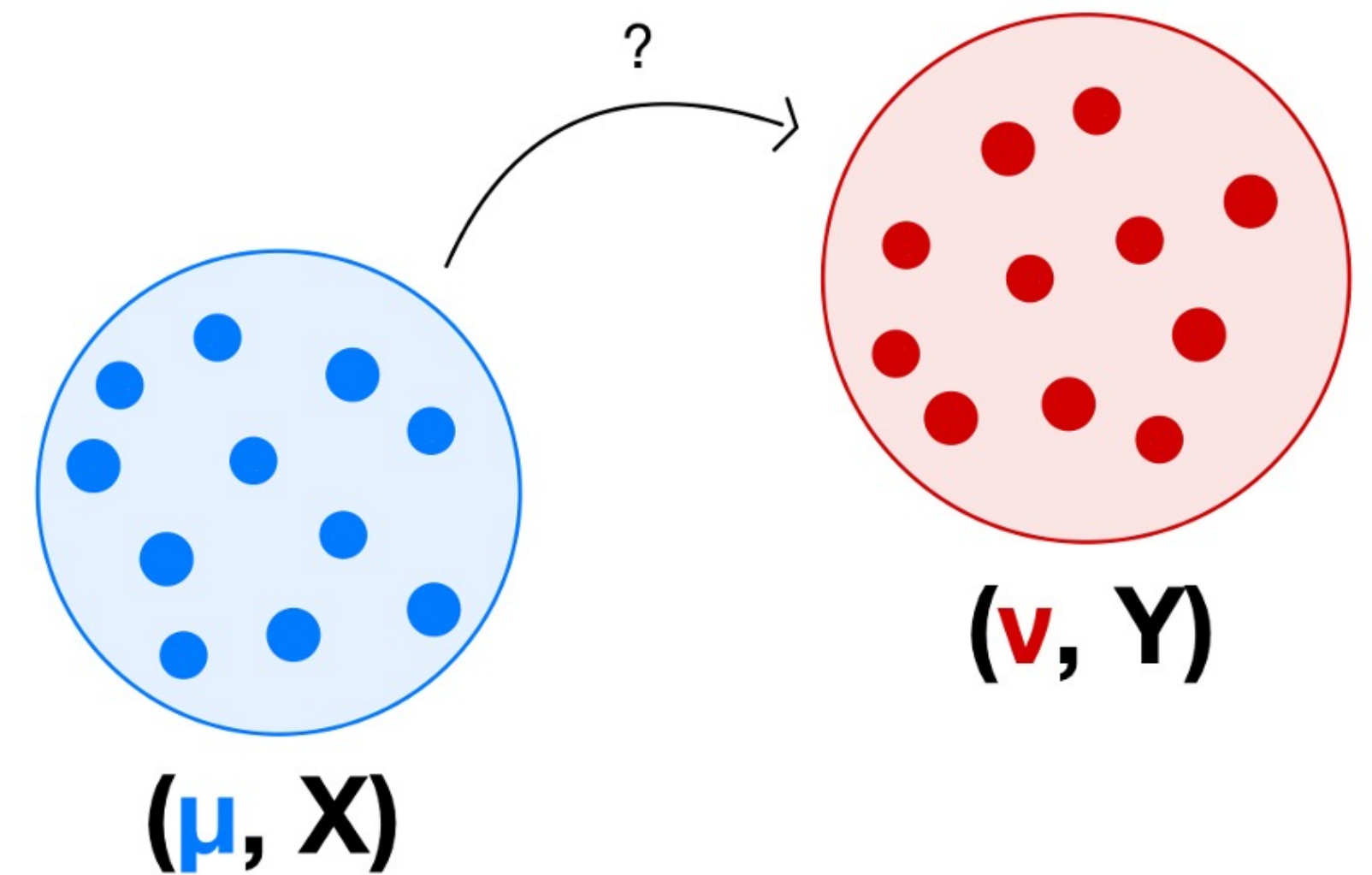
- $$L_C(\mu, \nu) := \min_{\pi \in \mathcal{U}(\mu, \nu)} \int_{X \times Y} c(x, y) d\pi(x, y)$$

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loss

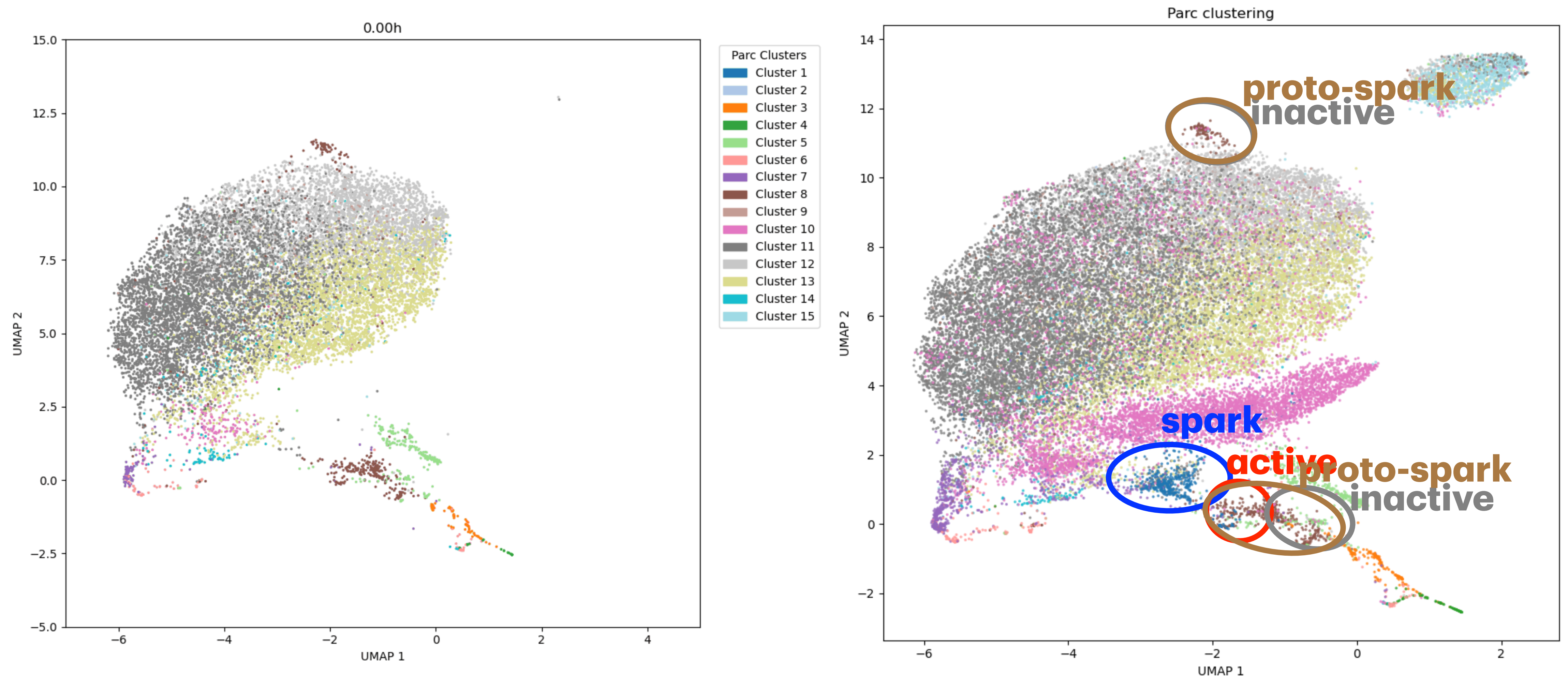
function

- Generalized “interpolation” for distributions
- Cells sitting close to each other in expression space are connected more than cells sitting far apart



Stochasticity in cancer immunotherapies

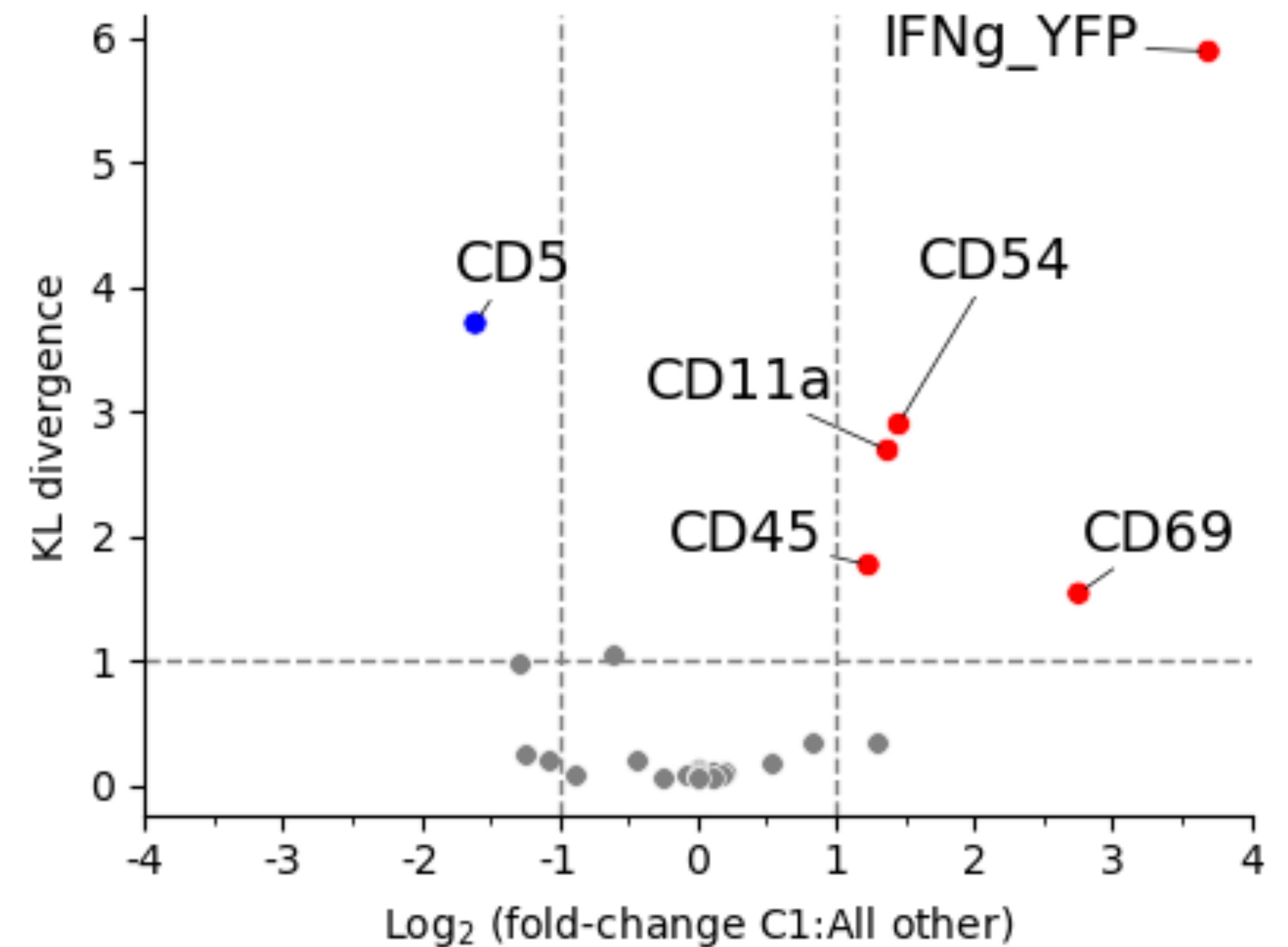
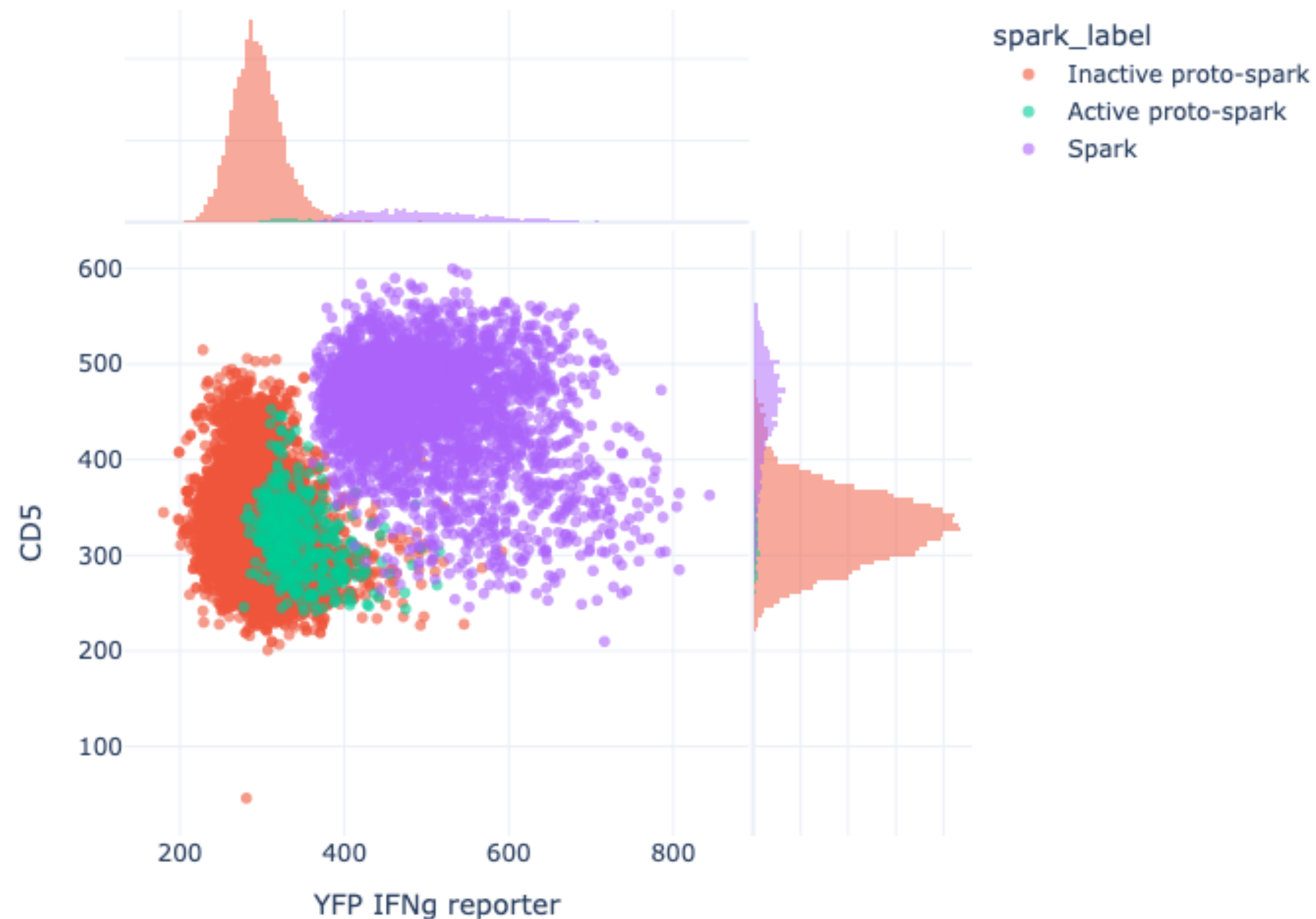
Optimal transport reliably uncovers rare mover cells



Stochasticity in cancer immunotherapies

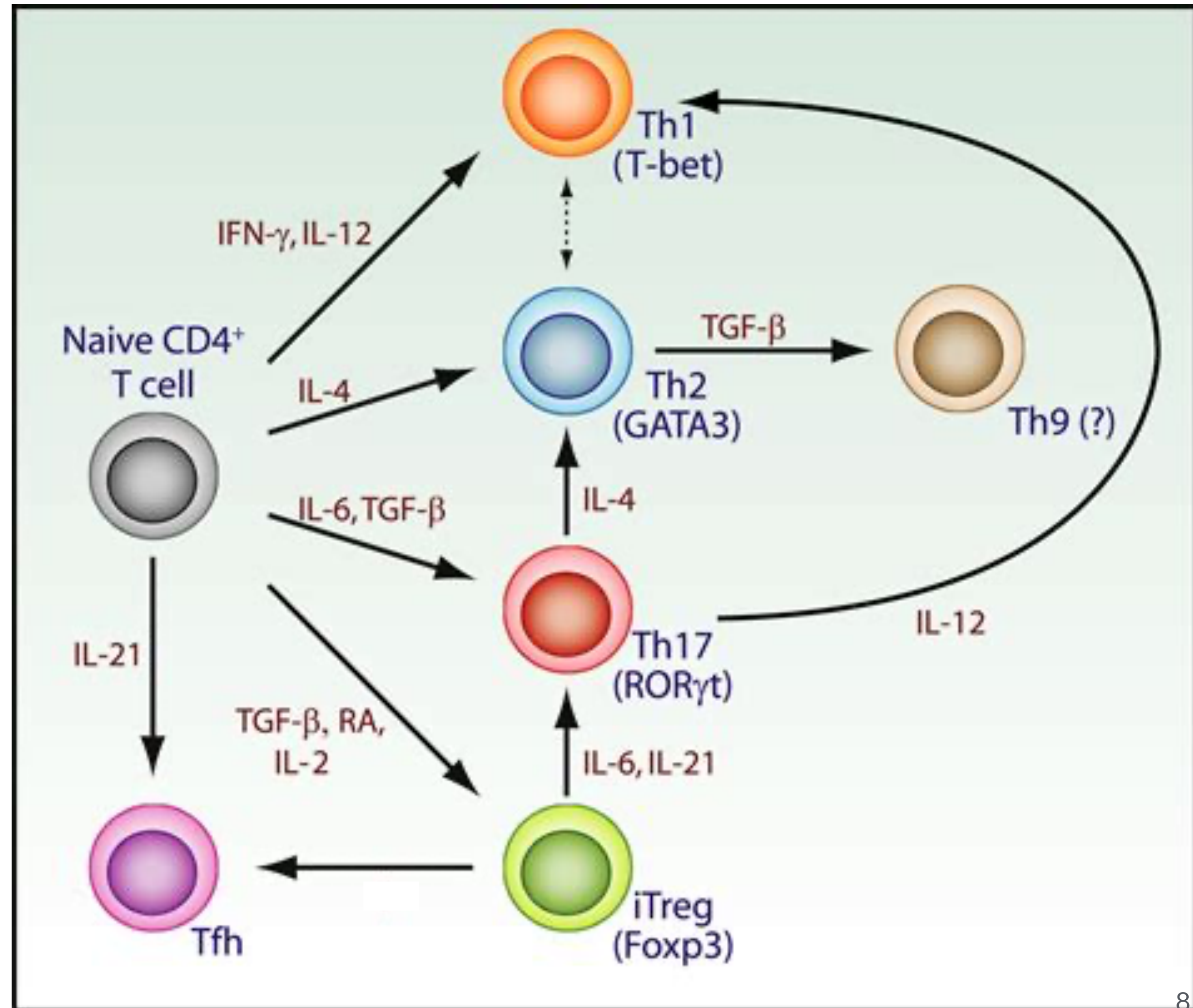
Realistic cell states lie on a continuum

Gate 1: YFP IFNg reporter vs CD5



Cells moving through the landscape change function

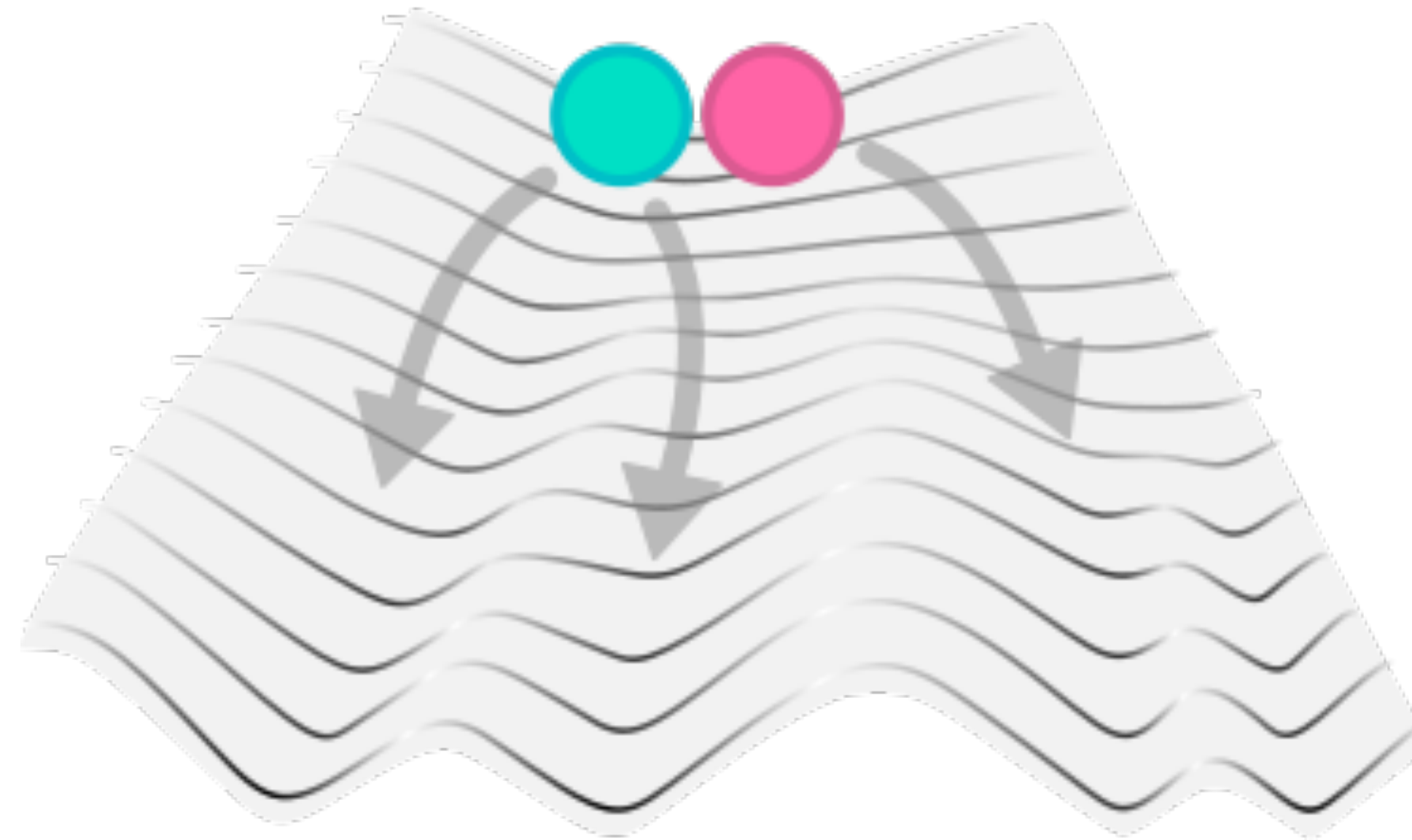
- Immune cells change their function during response
- The landscape can include
 - Extrinsic : cell-cell signaling environment
 - Epigenetic layer (chromatin accessibility, histone modifications)



Cell fate landscape

Waddington landscape model

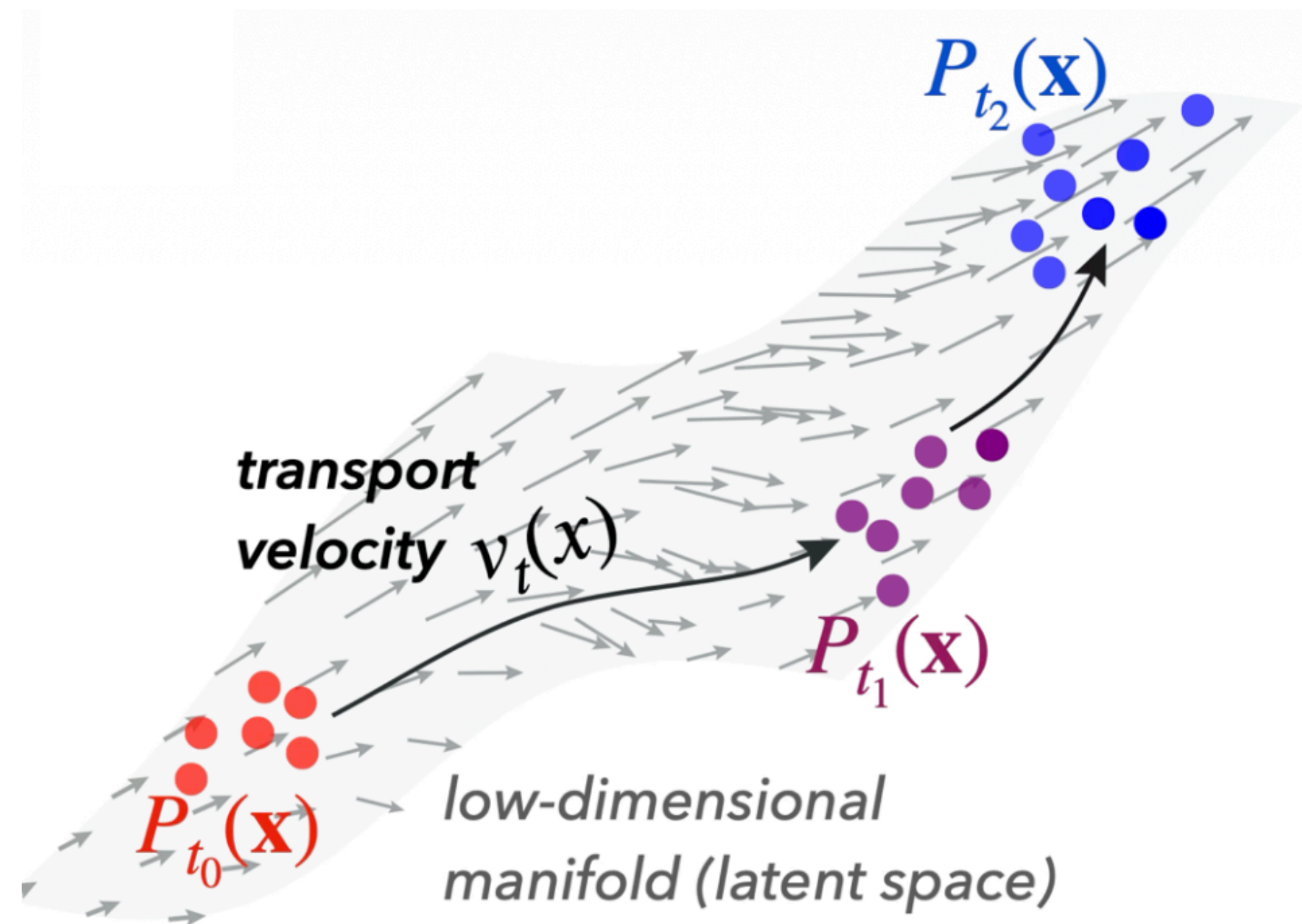
- Cells interact with a background potential => valleys correspond to stable cell states (e.g., activated, memory, IFN-g producer, etc.)



Dynamic optimal transport

Waddington landscape => seascape

$$\mathcal{W}_2^2[\rho_A, \rho_B] = \min_{\rho_s, \phi_s | s \in [0,1]} \left\{ \tau \int_0^\tau \int \|\mathbf{v}_s(x)\|^2 \rho_s(x) dx ds \mid \mathbf{v}_s(x) = \nabla \phi_s(x), \frac{\partial \rho_s}{\partial s} = \nabla \cdot (\rho_s \nabla \phi_s), \rho_0 = \rho_A, \rho_1 = \rho_B \right\}$$



$$T = \nabla \phi_t$$

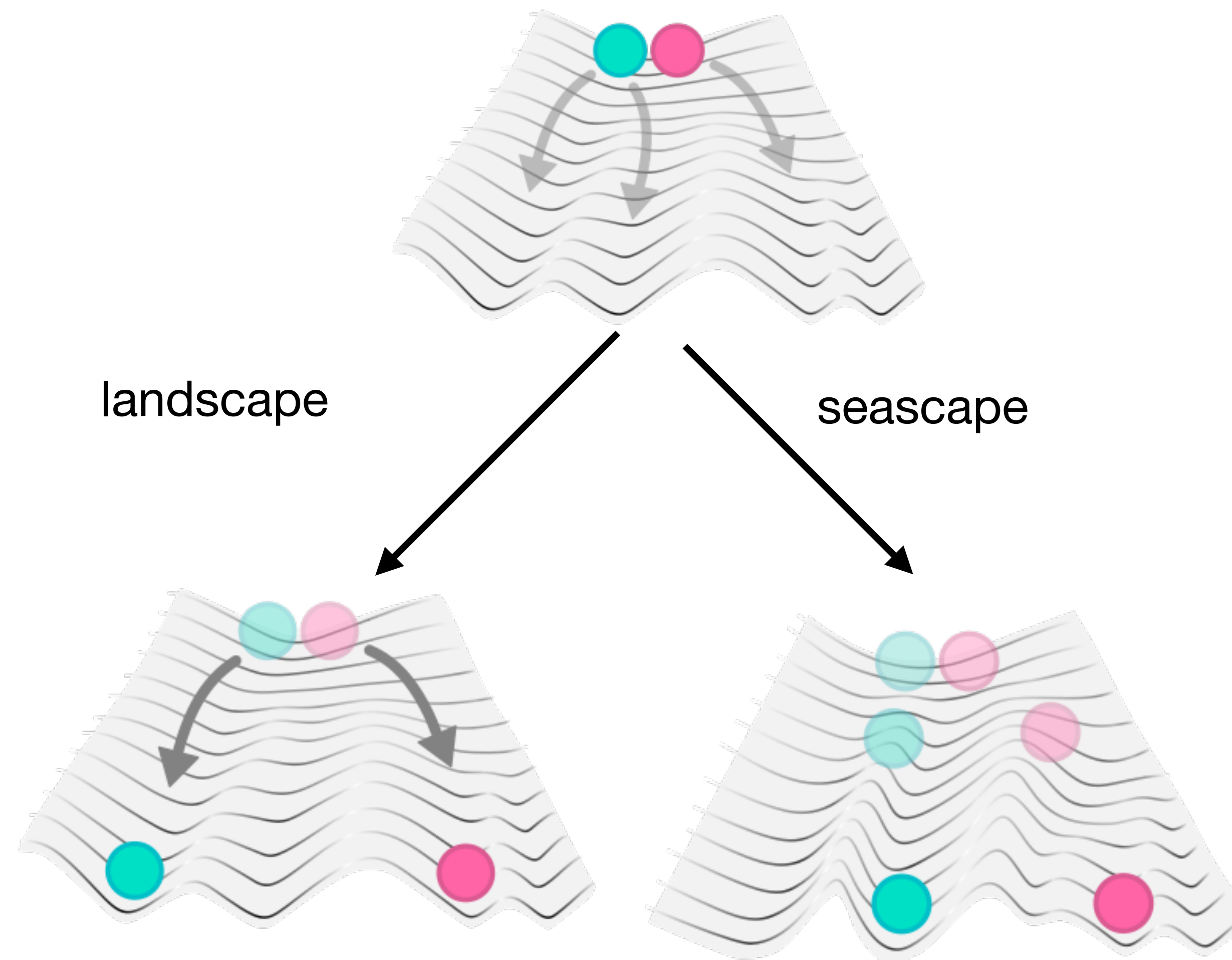
$$T_t(x) = (1 - t)x + tT(x), \quad t \in [0,1]$$

$$\rho_t = (T_t)_\# \rho_0$$

Constant speed geodesics

Cell fate landscape

Waddington landscape => seascape



Conclusion

- Functionally critical cells in cancer immunotherapies are identified using optimal transport
- Realistic cell states lie on a continuum
- Mapping out the continuous cell fate landscape (w/ optimal transport) will improve understanding of immune system dynamics
- Modeling individual cell contributions/perturbations to the global immune response is challenging

Thank you !

Nourmohammad Lab

